

Deliverable 2: The Code

Repository contents and how to run it

Mills-Operation · AI Reliability Copilot for Hot Mill Operations

Mills-Operation

An AI-powered reliability copilot for hot mill operations. Catches roll stand bearing degradation before it becomes unplanned downtime, and lets a shift supervisor ask a real AI copilot about the fleet in plain English.

CI failing

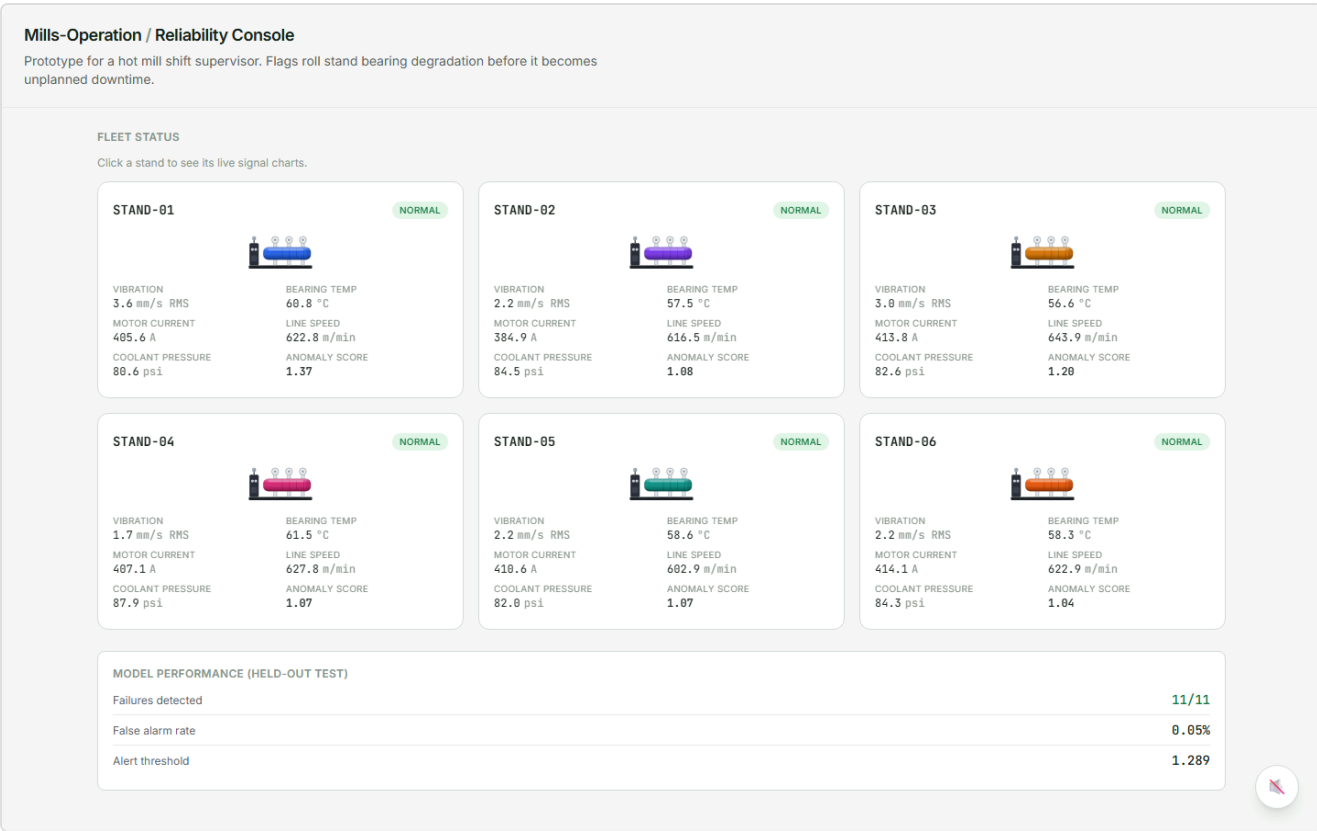
License MIT

Python 3.10+

React 19

TypeScript strict

AI LLM tool use



STAND-06

NORMAL

VIBRATION

2.2 mm/s RMS

MOTOR CURRENT

414.1 A

COOLANT PRESSURE

84.3 psi

BEARING TEMP

58.3 °C

LINE SPEED

622.9 m/min

ANOMALY SCORE

1.04

MODEL PERFORMANCE (HELD-OUT TEST)

Failures detected

False alarm rate

Alert threshold

11/11

0.05%

1.289

Overview

A shift supervisor watching a hot mill line needs one thing: to know a bearing is failing before it takes the line down. This project trains a real anomaly detector on realistic sensor signatures, runs it against a continuously ticking live feed, and wraps the whole thing in a console a supervisor would actually want to use, including an AI copilot they can ask real questions.

Every number on this dashboard is real output from a trained model, not a mockup. 11 out of 11 injected failures caught on held-out data, with a 0.05% false alarm rate. See [docs/architecture.md](#) for the full design story, including what didn't work the first time.

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What it does

Live Monitor	6 roll stands, ticking once every real minute, scored live by the trained detector. Filter by today, past week, last month, or a custom range reaching back through the training data.
Degradation Simulator	A virtual stand you control directly. Push any sensor toward failure by hand (the other four auto-correlate along the same curve real bearings follow), run a timed degradation, and watch the real model score it minute by minute.
Fleet Copilot	A real chat, not a canned FAQ. Ask it to compare stands, find who's alerting, check alert history, or predict who's trending toward trouble next, grounded in live data through 6 read-only tool calls, never invented.
Real detection, not a demo dataset	LocalOutlierFactor trained on a realistic non-linear bearing-wear signature, picked after an actual bake-off against 4 other models (see notebooks/model_comparison.ipynb), with an independent backstop for the model's own blind spot on sustained failures.

Screenshots

Live signal charts, with real alert history ![Stand detail with alert bands](docs/screenshots/stand-detail.png)	**Degradation simulator, live model response** ![Simulator mid-alert](docs/screenshots/simulator-alert.png)
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Fleet copilot, answering from real data

← Back to fleet

Mills-Operation / Reliability Console

Live monitorDegradation simulatorCopilot

COPILOT

Ask about any stand: compare readings, check who's alerting, look up alert history. Answers are grounded in real fleet data, never invented.

Clear chat

Explain STAND-01's status

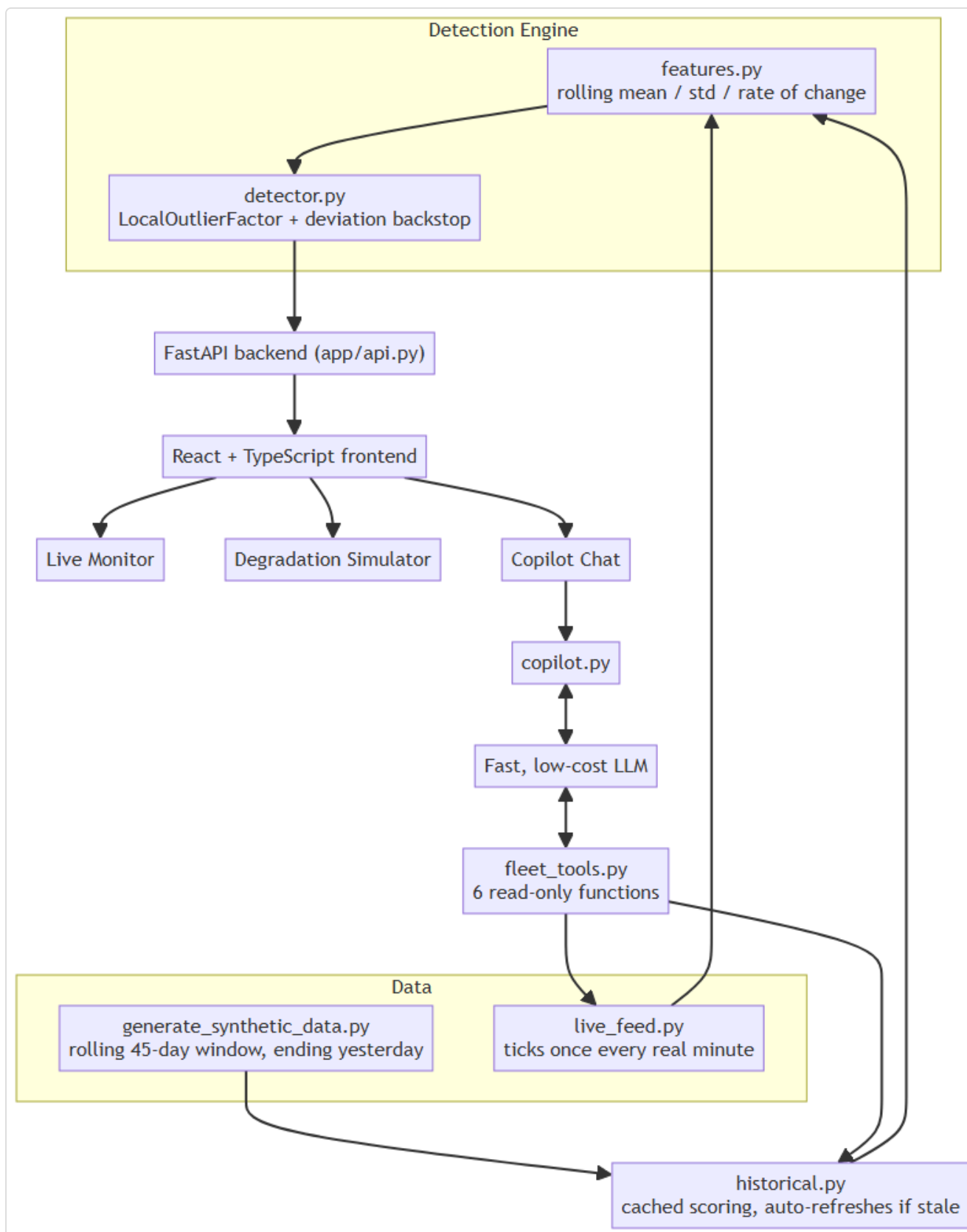
Which stand has the highest bearing temperature?

STAND-01 has the highest bearing temperature at 60.75°C.

Ask about the fleet...

Send

Architecture



The AI copilot doesn't use embeddings or a vector database. With only 6 stands, the entire fleet's current state trivially fits in a single prompt, so it's given 6 narrow, read-only Python functions instead (current reading, alert history, trend direction, and so on) and decides which ones a question actually needs. Full writeup in [docs/copilot-how-it-works.md](#).

Tech stack

Layer	Choice	Why
Data generation	Python, pandas, numpy	Deterministic, seeded, reproducible without a 25MB file in git history
Anomaly detection	scikit-learn, LocalOutlierFactor	Won an actual bake-off against IsolationForest, OneClassSVM, and EllipticEnvelope, not picked by reasoning alone
Backend	FastAPI	Serves scored data, runs the live feed as a background task, proxies the copilot
Frontend	React 19, TypeScript, Vite, Tailwind CSS v4	A real supervisor-facing tool, not a script re-running top to bottom on every click
Charts	Recharts	Alert bands as shaded regions, not thousands of individual markers
AI copilot	A fast, low-cost LLM, tool use	Fast and cheap for grounded, short explanations; tool use over RAG since the underlying data is small and structured
Testing	Playwright	Core flows gated in CI on every push
CI	GitHub Actions	Generate data, train, evaluate, test, gate on all of it

Full justification for every choice, including what was rejected and why, in [docs/stack-justification.md](#).

Getting started

```
# Python backend
pip install -r requirements.txt

# 1. Generate the synthetic sensor dataset (deterministic, ~389K rows, a few seconds)
python data/generate_synthetic_data.py

# 2. Train the anomaly detector and score the held-out test window
python -m app.evaluate

# 3. Start the backend API (also starts the live feed in the background)
uvicorn app.api:app --reload --port 8000
```

```
# React frontend, in a second terminal
cd web
npm install
npm run dev
```

Open `http://localhost:5173`. The frontend expects the API at `http://localhost:8000` by default (override with `VITE_API_URL` in `web/.env.local`).

Enabling the AI copilot

```
cp .env.example .env # then put a real ANTHROPIC_API_KEY in .env
```

Without a key, both the single-stand explanation and the fleet chat still work, they fall back to a deterministic, data-grounded response instead of a live model call, so the dashboard never breaks because of a missing key or a network blip.

Testing

```
npm install
npx playwright install --with-deps chromium
npx playwright test
```

Playwright builds and starts both the backend and the frontend itself (see `playwright.config.js`), so just make sure the two setup steps above have been run first. Runs automatically on every push and PR via `.github/workflows/ci.yml`.

Project structure

```
app/      anomaly detection engine, live feed, simulator, AI copilot, FastAPI backend
web/      React + TypeScript frontend
tests/    Playwright test suite
data/     synthetic dataset generator
notebooks/ model comparison bake-off
docs/     architecture, stack justification, AI partnership log, data strategy, security & risk
scripts/  one-off utilities (README screenshot capture)
```

Documentation

Doc	What's in it
docs/architecture.md	System design, key decisions, what breaks at Nucor scale
docs/copilot-how-it-works.md	How the AI copilot actually works, in plain English
docs/stack-justification.md	Why this stack, and why not C3.ai or Palantir Foundry
docs/ai-partnership-log.md	How AI was used building this, including every override and every real bug it took to get here
docs/data-strategy.md	The synthetic data approach, and what real data this would need in production
docs/security-risk.md	Risk assessment, including the AI copilot's specific guardrails

Known limitations

- One global anomaly threshold across all 6 stands, not per-stand calibration
- Trained and evaluated entirely on synthetic data; see [docs/data-strategy.md](#) for what's unverified
- Targets one failure mode (roll stand bearing degradation) by design, not general downtime prediction
- The live feed is a single in-process background task with an in-memory buffer, not production-scale streaming

Full breakdown, including what this would take to fix at scale, in [docs/architecture.md](#) .

Built as a take-home assignment, doubling as a demonstration of AI-assisted engineering: real ML evaluation, a real bug list, and an AI partnership log that says what actually happened, not what looks good.